

Yulin Xia

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Skills

Programming Python(5 yrs), C/C++(4 yrs), Java, Bash, Go, TypeScript, JavaScript.
AI & Perception Computer Vision, Machine Learning, Pose Estimation, PyTorch, OpenCV, NumPy, Jupyter, Kubeflow.
Systems Linux, Docker, AWS, Kubernetes, Terraform, GitHub Actions, Datadog, Git, CI/CD.

Education

University of Waterloo Waterloo, ON
Bachelor in Computer Engineering Sept 2022 - Current

Work Experience

TextNow Waterloo, Canada
Site Reliability Engineer Jan 2026 - Apr 2026

- Added **AI agent** to **GitHub Actions Terraform** plan scans: parses drift/errors into structured summaries (error types, top offenders, pass rates), proposes fixes before apply; engineers review prod changes.
- Fixed **Terraform** drift and **IAM** gaps; reconciled **AWS** state, tightened deploy-role permissions; updated deployment runbooks with rollback and on-call handoff steps.
- Hardened **S3/TLS/VPC**, security groups, and **Route 53**; tuned **Datadog**, **Kibana**, and SMS for cost-aware alerts. Removed unused LBs after **Kubernetes/AWS** checks via **Terraform/kubectrl**; logged capacity and audit results.

TextNow Waterloo, Canada
Site Reliability Engineer May 2025 - Aug 2025

- Modernized and maintained **AWS** infrastructure using **Terraform** and **Atlantis**; updated legacy **EC2** kernel nodes, ENIs, NLBs, and **Route 53** DNS records across multiple environments.
- Deployed full-stack JTL Reporter service using **YAML** and **ECR** repositories; set up **Aurora PostgreSQL**, **IRSA**, **Helm**-based Nexus HA deployment, and completed cross-service SQL migrations.
- Enhanced cross-account **S3**; AWS Health to Slack; **EKS/Istio** upgrades; **SageMaker** for data science via **IAM/VPC**.

Ford Motor Company Waterloo, Canada
Software Developer, SYNC HMI Sep 2024 - Dec 2024

- Developed central SYNC **React/JSX** settings components to improve in-vehicle HMI efficiency on the head-unit.
- Created custom **React hooks** to reduce UI duplication and improve settings-screen performance.
- Ran **unit tests** on hooks and components; validated on **vehicle hardware** before control-screen handoff.

Ford Motor Company Waterloo, Canada
Software Developer, SYNC HMI Jan 2024 - Apr 2024

- Developed the SYNC HMI system with advanced analytics in **climate control**, **Menlo** settings, and **Menlo media**, tracking user modifications and tab access after software releases.
- Conducted **Linux** system modifications; ran **Jenkins** tests and ensured integration before each vehicle release; code reviews improved coding standards and reliability before the release branch.

Caerulean AI Toronto, Canada
Software Engineer May 2023 - Aug 2023

- Used **TypeScript** and **React** to develop **7** frontend pages; designed and implemented **REST APIs**; used **Prisma** for database changes and **Docker** for local database setup mirroring production.
- Designed and deployed a **CI/CD** pipeline using **EKS** and **Argo** use **uff** ow **ovg** a **scpline** e